

# ARI-SPEC

ARI — Agent Runtime Interface

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## ARI — Agent Runtime Interface

**Version:** 0.1 (Draft) **Status:** Draft — Request for Comments

**Reference implementation:** [marrow](#) **License:** Apache-2.0 (spec text under CC BY 4.0 — see §1.3)

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### Abstract

The **Agent Runtime Interface (ARI)** is a language-neutral contract for the *runtime layer* of an LLM agent system: the loop that holds conversation state, calls a model provider, dispatches tools, routes messages between agents, and manages cancellation, timeouts, persistence, and observability.

ARI does **not** standardize *what* tools exist, *how* a model is prompted, or *how* agents talk to each other across a network — those layers are owned by other specifications (notably **MCP** and **A2A**; see §1.3). ARI standardizes the layer *beneath* them: the embeddable engine that actually runs an agent turn.

A runtime that implements this contract is **ARI-conformant** and can be substituted for any other ARI-conformant runtime without changing application code that targets the interface. [marrow](#) is the reference implementation.

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## 1. Introduction

### 1.1 Motivation

Today's agent frameworks (LangGraph, CrewAI, AutoGen) are excellent and mature, but they share an architectural assumption: **the runtime is written in Python and runs on a server with a network path to a hosted model**. That assumption fails in a growing class of deployments:

- **Embodied / robotic** systems with hard real-time constraints, where garbage-collection pauses and a heavyweight interpreter on the hot path are unacceptable.
- **Edge / on-device** deployments on resource-constrained hardware, often with a local model and no reliable network.
- **Native applications** (game engines, audio/DSP, trading systems, CAD) that need to embed an agent loop without shipping a Python runtime to the customer.

These deployments cannot adopt a Python-locked runtime. Yet there is no neutral contract that says *what an agent runtime must expose* — so every native re-implementation reinvents the surface, and nothing is interoperable.

ARI fills that gap. It defines the contract once, so that:

- A native runtime (C++, Rust, Zig, ...) can claim conformance and be a drop-in.
- Tooling (tracers, state stores, provider adapters) can target the contract rather than a specific framework.
- "Agents you can't ship Python into" becomes a portable, standardized target.

## 1.2 Scope

**ARI standardizes:**

| Area                                                         | Section             |
|--------------------------------------------------------------|---------------------|
| Message / role data model                                    | <a href="#">§2</a>  |
| Model provider interface (generate, stream, cancel, timeout) | <a href="#">§3</a>  |
| Agent conversation state                                     | <a href="#">§4</a>  |
| Tool registration and invocation contract                    | <a href="#">§5</a>  |
| Multi-agent routing (send, drain, handoff, backpressure)     | <a href="#">§6</a>  |
| Runtime lifecycle (shutdown)                                 | <a href="#">§7</a>  |
| State persistence                                            | <a href="#">§8</a>  |
| Observability (trace spans)                                  | <a href="#">§9</a>  |
| Conformance profiles                                         | <a href="#">§10</a> |

**ARI explicitly does NOT standardize:**

- The wire protocol for tool *definitions* or context exchange — that is **MCP**.
- Cross-process / cross-network agent-to-agent messaging — that is **A2A**.
- Prompt formats, model weights, or inference engines.
- Any specific programming language's binding ergonomics (those are implementation choices; see [Appendix A](#)).

## 1.3 Relationship to MCP and A2A

ARI is **complementary, not competitive**, with the emerging agent protocol stack:

|                |                                                                               |  |
|----------------|-------------------------------------------------------------------------------|--|
| A2A            | – agent-to-agent messaging (across hosts)                                     |  |
| MCP            | – tool / context exchange (model ↔ world)                                     |  |
| ARI            | – the runtime that runs a turn ← here<br>(state · provider · tools · routing) |  |
| Model provider | (OpenAI / Anthropic / local)                                                  |  |

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An ARI runtime SHOULD be able to expose its tools over MCP and participate in an A2A mesh, but neither is required for ARI conformance. ARI is the layer you embed; MCP and A2A are protocols you speak.

## 1.4 Terminology

The key words **MUST**, **MUST NOT**, **REQUIRED**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **MAY**, and **OPTIONAL** in this document are to be interpreted as described in [RFC 2119](#).

- **Runtime** — an implementation of this interface.
- **Agent** — a named, stateful conversation participant within a runtime.
- **Provider** — an adapter that turns a request into a model completion.
- **Turn / Step** — one provider invocation that appends to agent state.
- **Host application** — the code embedding the runtime.

## 1.5 Conformance overview

Conformance is claimed against a **profile** (§10). Every conformant runtime MUST satisfy **ARI-Core**. Runtimes targeting embedded or server deployments MAY additionally satisfy **ARI-Embedded** or **ARI-Server**. A runtime MUST NOT advertise "ARI-conformant" without naming the profile(s) it passes and the version of the conformance suite used.

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# 2. Data Model

## 2.1 Role

A message role MUST be one of exactly four values:

| Role      | Meaning                                                 |
|-----------|---------------------------------------------------------|
| system    | Instructions that condition the agent.                  |
| user      | Input from outside the agent.                           |
| assistant | Output produced by the agent / model.                   |
| tool      | The result of a tool invocation, fed back into context. |

Runtimes MUST NOT define additional roles in the core data model. Vendor extensions belong in message metadata (§2.2).

## 2.2 Message

A **Message** is the atomic unit of conversation state. It MUST carry:

| Field        | Type         | Required | Notes                             |
|--------------|--------------|----------|-----------------------------------|
| role         | Role         | yes      | See §2.1.                         |
| content      | UTF-8 string | yes      | The message body. May be empty.   |
| name         | string       | no       | Author/tool name; empty if unset. |
| timestamp_ms | int64        | no       | Unix epoch milliseconds; 0 if     |

|          |                    |    |                                             |
|----------|--------------------|----|---------------------------------------------|
| metadata | map<string,string> | no | unset.<br>Opaque key/value extension point. |
|----------|--------------------|----|---------------------------------------------|

Runtimes MUST treat metadata as opaque pass-through and MUST preserve it across persistence and routing. Unknown keys MUST NOT cause errors.

## 2.3 Content limits

A runtime MUST enforce a maximum content size and MUST reject (not truncate) messages exceeding it, surfacing an error to the caller. The default ceiling is **4 MiB**. Runtimes targeting **ARI-Embedded** (§10.2) MAY set a lower fixed ceiling but MUST document it. This protects against out-of-memory via runaway tool output.

# 3. Provider Interface

A **Provider** turns a `GenerationRequest` into a completion. Providers are the ARI extension point for model vendors and local inference engines.

## 3.1 GenerationRequest

| Field        | Type        | Required | Default | Notes                             |
|--------------|-------------|----------|---------|-----------------------------------|
| model        | string      | yes      | —       | Provider-specific model id.       |
| messages     | Message[]   | yes      | —       | Ordered conversation context.     |
| temperature  | float       | no       | 0.7     | Sampling temperature.             |
| max_tokens   | int         | no       | 1024    | Upper bound on completion length. |
| timeout_ms   | int         | no       | 0       | Wall-clock timeout; 0 disables.   |
| cancel_token | CancelToken | no       | none    | Cooperative cancellation handle.  |

## 3.2 GenerationResponse

| Field             | Type   | Required | Notes                              |
|-------------------|--------|----------|------------------------------------|
| content           | string | yes      | The completion text.               |
| prompt_tokens     | int    | no       | 0 if the provider does not report. |
| completion_tokens | int    | no       | 0 if the provider does not report. |

Token counts are best-effort. A provider that cannot report usage MUST return 0, never a fabricated estimate.

## 3.3 generate (synchronous)

A provider **MUST** implement a synchronous `generate(request) -> response`. It **MUST** honor `cancel_token` at natural yield points and **SHOULD** honor `timeout_ms`; if it does not honor the timeout internally, the runtime **MUST** enforce it externally (e.g. by running the call on a worker and abandoning it). On cancellation or timeout the provider/runtime **MUST** surface a distinguishable error, not a partial success.

### 3.4 generate\_stream (streaming)

A provider **SHOULD** implement `generate_stream(request, on_chunk)`, invoking the callback once per chunk such that concatenating all chunks reconstructs the full content. A provider that cannot stream **MUST** fall back to emitting the entire content as a single chunk so that callers written against the streaming API still work. Conformance to streaming is part of **ARI-Server**; it is **OPTIONAL** in **ARI-Core**.

### 3.5 Cancellation

A runtime **MUST** provide a **CancelToken** primitive supporting:

- `cancel()` — request cancellation (idempotent).
- `cancelled()` — non-blocking check.
- `reset()` — clear state for reuse.

Cancellation is cooperative: long-running providers and tools **MUST** poll `cancelled()` at yield points. `CancelToken` operations **MUST** be safe to call from a thread other than the one running the request.

### 3.6 Timeouts

`timeout_ms` is wall-clock. A value of 0 disables the timeout. When a timeout elapses the runtime **MUST** stop waiting and surface a timeout error; it **SHOULD** also trip the request's `cancel_token` so cooperating providers can release resources.

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## 4. Agent State

An **AgentState** holds the ordered message history and optional system prompt for one agent. It **MUST** expose:

| Operation                                              | Contract                                                          |
|--------------------------------------------------------|-------------------------------------------------------------------|
| <code>id()</code>                                      | Stable, immutable agent identifier.                               |
| <code>append(message)</code>                           | Append to history; <b>MUST</b> be safe under concurrent readers.  |
| <code>history()</code>                                 | Snapshot of all messages in order.                                |
| <code>size()</code>                                    | Current message count.                                            |
| <code>clear()</code>                                   | Drop all messages (system prompt retained unless re-set).         |
| <code>set_system_prompt(text) / system_prompt()</code> | Set/get the conditioning prompt.                                  |
| <code>trimmed(n)</code>                                | Return at most the most-recent <code>n</code> messages, in order. |

State reads **MUST** be consistent: a `history()` snapshot **MUST NOT** observe a partial append. Implementations are **RECOMMENDED** to use read-heavy synchronization (e.g. a shared/reader-writer lock) since history is read far more than written.

trimmed(n) MUST preserve chronological order and MUST NOT drop a system message that the runtime injects separately; trimming applies to conversational history, and the system prompt is prepended at request-build time.

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## 5. Tool Interface

### 5.1 Tool descriptor

A tool MUST be registered with:

| Field       | Type     | Required | Notes                          |
|-------------|----------|----------|--------------------------------|
| name        | string   | yes      | Unique within a registry.      |
| description | string   | no       | Human/model-facing summary.    |
| schema_json | string   | no       | JSON Schema of the arguments.  |
| fn          | callable | yes      | The implementation (see §5.2). |

The `schema_json` field is a **string** containing JSON Schema, stored verbatim. ARI does not mandate how the schema is produced (hand-written, generated from type annotations, etc.).

### 5.2 Invocation contract

The tool implementation is a pure **JSON-in / JSON-out** function:

```
invoke(name: string, args_json: string) -> result_json: string
```

- `args_json` MUST be a JSON object string.
- The runtime MUST raise a distinguishable error if `name` is not registered.
- The implementation language is irrelevant to the contract: the reference implementation keeps the registry and dispatch in C++ while tool bodies may be written in the host language.

### 5.3 Result envelope

A tool invocation result MUST be a JSON object using this envelope:

```
{ "ok": true, "result": <any JSON value> }
{ "ok": false, "error": "<message>" }
```

On failure, the runtime MUST set `"ok": false` and MUST place a human-readable string in `"error"`. The runtime MUST NOT leak internal stack traces or host exception details into `"error"` (see §5.4).

### 5.4 Limits and redaction

A runtime MUST support configurable size caps on both `args_json` and the result, and MUST redact internal error detail by default. These limits protect the context window and prevent information disclosure. **ARI-Embedded** runtimes MUST enforce a fixed, documented cap.

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## 6. Multi-Agent Routing

A runtime MUST provide a **Router** mediating messages between named agents.

### 6.1 send / drain

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| Operation                                 | Contract                                                  |
|-------------------------------------------|-----------------------------------------------------------|
| register_agent(id) / unregister_agent(id) | Manage routable identities.                               |
| send(from, to, message)                   | Enqueue a message into to's inbox.                        |
| drain(id)                                 | Atomically remove and return all pending messages for id. |

drain MUST be atomic with respect to concurrent send calls: a drained message MUST NOT also remain in the inbox, and a message MUST NOT be lost between two drains.

## 6.2 handoff and active agent

A runtime MUST support handoff(from, to, seed?) — transfer of control with an optional seed message — and MUST track an OPTIONAL "active" agent (active() / set\_active(id)). Handoff to an unregistered agent MUST fail without mutating state.

## 6.3 Backpressure

A runtime MUST allow a per-agent inbox bound and an overflow policy:

| Policy     | Behavior on full inbox                                      |
|------------|-------------------------------------------------------------|
| Reject     | Throw/return error — backpressure surfaced to the producer. |
| DropOldest | Evict the oldest pending message, accept the new one.       |
| DropNewest | Silently discard the incoming message.                      |

The default bound MAY be unbounded (0) for **ARI-Server**, but **ARI-Embedded** runtimes MUST default to a finite bound to guarantee bounded memory.

## 7. Lifecycle

A runtime MUST support graceful shutdown:

- shutdown() — signal that no new top-level work should start. Idempotent.
- is\_shutdown() — non-blocking status check.

After shutdown(), operations that take ownership of new resources (e.g. creating an agent) MUST fail. In-flight work in other threads MUST be allowed to complete; shutdown() MUST NOT forcibly abort running turns. Forcible termination, if offered, MUST be a separate, explicitly-named operation.

## 8. Persistence

Persistence is OPTIONAL for **ARI-Core** and REQUIRED for **ARI-Server**. A runtime that offers persistence MUST do so behind a **StateStore** contract that supports, by behavior:

| Operation             | Contract                                                                     |
|-----------------------|------------------------------------------------------------------------------|
| save(agent_id, state) | Durably store a snapshot of the agent's history (and system prompt, if any). |
| load(agent_id)        | Return the stored messages, or an explicit <b>absent</b> result if none.     |

|                                                    |                            |
|----------------------------------------------------|----------------------------|
| <code>delete(agent_id)</code>                      | Remove a stored snapshot.  |
| <code>enumerate</code> (e.g. <code>keys()</code> ) | List the stored agent ids. |

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A round-trip **MUST** be **lossless** for all **REQUIRED** Message fields *and* metadata (§2.2). A load of an unknown `agent_id` **MUST** return an explicit "absent" result (e.g. `null/None`), not an error. At least one durable backend (e.g. SQLite) **SHOULD** be provided; an in-memory backend **MAY** be provided for testing. A runtime **MAY** additionally offer a convenience that rehydrates a live agent directly from a store; the signature of such a helper is an implementation detail, not part of this contract.

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## 9. Observability

Observability is **OPTIONAL** for **ARI-Core** and **RECOMMENDED** elsewhere. A runtime that emits traces **MUST** do so behind a **TraceSink** contract that exposes a `span(name, attributes)` operation returning a span handle supporting:

- `set_status(ok: bool)`
- `record_exception(error)`
- `scope entry/exit` (a span has a clear begin and end).

Runtimes **SHOULD** provide a no-op sink (zero overhead when disabled) and **SHOULD** offer an OpenTelemetry-compatible sink. A disabled **TraceSink** **MUST** add no measurable allocation on the hot path — this is a hard requirement for **ARI-Embedded**.

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## 10. Conformance Profiles

A runtime claims conformance to one or more **profiles**. Profiles are additive: **ARI-Embedded** and **ARI-Server** both presuppose **ARI-Core**.

### 10.1 ARI-Core

The mandatory baseline. A runtime is **ARI-Core conformant** if it satisfies:

- §2 Data model (all **REQUIRED** fields, 4 roles, content cap)
- §3.1–3.3, §3.5–3.6 Provider generate, cancellation, timeouts
- §4 Agent state
- §5 Tool registration, invocation, result envelope, limits
- §6 Routing (send/drain/handoff) and backpressure policies
- §7 Graceful shutdown

Streaming, persistence, and observability are **OPTIONAL** at this level.

### 10.2 ARI-Embedded

The profile this specification is built to serve: agent runtimes embedded in **robotic, edge, and native** hosts. In addition to **ARI-Core**, an **ARI-Embedded** runtime **MUST**:

1. **Be embeddable without a managed-language runtime.** The core **MUST** compile/run as a native library with no required dependency on a garbage-collected interpreter. (Bindings to such languages **MAY** exist; they **MUST NOT** be required.)
2. **Bound its memory.** All queues/inboxes **MUST** default to a finite bound (§6.3); content and tool I/O caps **MUST** be fixed and



documented (§2.3, §5.4).

3. **Add no hot-path allocation when observability is disabled** (§9).
4. **Support cooperative cancellation and wall-clock timeouts** as first-class, thread-safe primitives (§3.5–3.6).
5. **Declare its dependency footprint.** The minimal build MUST document every third-party dependency; the target is "near-zero."

**Why this profile is the differentiator:** a Python-locked framework cannot satisfy requirements 1 and 3 without ceasing to be Python-locked. ARI-Embedded defines a lane the incumbent runtimes structurally cannot enter.

## 10.3 ARI-Server

The profile for high-scale, server-side orchestration. In addition to ARI-Core, an **ARI-Server** runtime MUST satisfy §3.4 (streaming), §8 (persistence with a durable backend), and §9 (observability with an OTel-compatible sink), and SHOULD support concurrent turns with real parallelism (e.g. releasing interpreter locks during provider I/O).

## 10.4 Claiming conformance

A runtime claiming ARI conformance MUST:

1. Name the profile(s) it passes (e.g. "ARI-Core + ARI-Embedded").
2. State the spec version (e.g. "ARI 0.1") and conformance-suite version.
3. Publish its results from the **ARI Conformance Kit** (see [Appendix B](#) and the strategy document).

The phrase "ARI-compatible" without a named profile and version is meaningless and MUST NOT be used in distribution materials.

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## 11. Versioning and Stability

- This spec uses **semantic versioning**. Within a major version, changes MUST be additive (new OPTIONAL fields, new profiles) — never a silent tightening of an existing MUST.
  - A field's REQUIRED/OPTIONAL status MUST NOT change within a major version.
  - Deprecations MUST be announced one minor version before removal and MUST carry a documented migration path.
  - The reference implementation's public API stability is governed by its own STABILITY.md; spec stability and implementation stability are versioned independently.
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## 12. Security Considerations

- **Untrusted tool output** is the primary injection vector. Runtimes MUST apply content (§2.3) and result (§5.4) caps, and MUST redact internal error detail.
- **Resource exhaustion** via unbounded inboxes is mitigated by mandatory bounds in ARI-Embedded (§6.3).
- **Cancellation/timeout** are security primitives, not just ergonomics: they bound the blast radius of a hostile or hung provider/tool.
- **Persistence** stores conversation content; runtimes MUST document the trust boundary of any StateStore backend.

## 13. Licensing of this document

The ARI specification text is offered under **CC BY 4.0** so that any party may implement it freely and reproduce the contract. The reference implementation (marrow) is licensed separately under Apache-2.0. Implementing ARI requires no license to the marrow code.

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## Appendix A: Reference bindings

ARI is language-neutral; the reference implementation provides two bindings that illustrate the contract. These are **informative**, not normative — a conformant runtime need not match these signatures, only the behavior in §§2-10.

**C++ (native core):** Provider, AgentState, AgentRouter, ToolRegistry, Engine, CancelToken, Message, GenerationRequest/Response — see [src/core/engine.hpp](#).

**Python (host binding):** Agent, Runtime, tool/ToolBox, Graph, StateStore, TraceSink, AsyncRuntime — see [python/marrow/](#).

## Appendix B: Conformance checklist

A runtime is **ARI-Core conformant** when every box is checked. ARI-Embedded and ARI-Server add their own checklists (maintained alongside the Conformance Kit).

- ☐ Four roles, no more (§2.1)
- ☐ Message carries all REQUIRED fields; metadata preserved through routing + persistence (§2.2)
- ☐ Content cap enforced by rejection, not truncation (§2.3)
- ☐ generate honors cancel token; timeout enforced internally or by runtime (§3.3, §3.6)
- ☐ CancelToken is thread-safe and idempotent (§3.5)
- ☐ AgentState reads are consistent under concurrent append (§4)
- ☐ Tool invocation uses the JSON envelope; unknown tool errors distinguishably (§5.2-5.3)
- ☐ Tool arg/result caps configurable; internal errors redacted (§5.4)
- ☐ drain is atomic vs concurrent send (§6.1)
- ☐ All three overflow policies implemented (§6.3)
- ☐ shutdown is idempotent; blocks new work; lets in-flight finish (§7)

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*ARI 0.1 (Draft). Reference implementation: marrow. Comments and proposed changes: open an issue tagged `ari-spec`.*